Grade: Check Plus

1. EXPECTED VALUES AND OPTIONS

A stock has a current selling price of $20. The distribution of the price of the stock one year from now is believed to be as follows:

PRICE (x)	PROBABILITY f(x)
20	.05
21	.06
22	.07
23	.08
24	.09
25	.12
26	.15
27	.13
28	.10
29	.10
30	.05
	1.00

An investor holds a one year option (European) on this stock. This enables him to purchase, one year from now, one share of stock at $25 per share. The value of his option will depend upon the price of the stock one year from now. If the price rises to $28, the value of the option (ignoring interest and commissions) is $3. If the price rises to only $23, the option is worthless.

1. Compute the mean and standard deviation of the stock price.
2. a. What is the probability that the stock price will fall within 1 standard deviation of the mean one year from now?
   b. What is the probability that the stock price will fall within 2 standard deviations of the mean one year from now?
3. Derive the distribution of the option value, ignoring interest and commissions.
4. What is the probability that the option will have positive value?
5. Compute the mean and standard deviation of the value of the option.

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Problem Set due Feb 24th

2. Expected Value and Options

x	f(x)	μ	σ^2	v	g(v) = P(V=v)
20	.05	1	1.46	0	
21	.06	1.26	1.17	0	
22	.07	1.54	0.81	0	
23	.08	1.84	0.46	0	
24	.09	2.16	0.18	0	
25	.12	3	0.02	0	.47
26	.15	3.9	0.05	1	.15
27	.13	3.51	0.33	2	.13
28	.10 ✓Q1	2.8	0.67	3	.10
29	.10	2.9	1.29	4	.10
30	.05	1.5	1.05	5	.05
	1.00	25.41	7.5		1.00

$\mu = x f(x)$

$\sigma^2 = \sum (x-\mu)^2 f(x)$

$\sigma = \sqrt{\sigma^2}$

1.) $\mu = 25.41$

$\sigma = 2.74$

2a.) $\mu - \sigma = 22.67$

$\mu + \sigma = 28.15$

$22.67 \le x \le 28.15$

$23 \le x \le 28$

23, 24, 25, 26, 27, 28

.08 + .09 + .12 + .15 + .13 + .10 = .67

Prob = 67% ✓Q2a within 1 std dev

2b.) $\mu - 2\sigma = 19.93$

$\mu + 2\sigma = 30.89$

$19.93 \le x \le 30.89$

$20 \le x \le 30$

↳ all values are within this range

Probability = 100% ✓Q2b within 2 std. dev

3.)

value (v)	g(v)
0	.47
1	.15
2	.13
3	.10 ✓Q3
4	.10
5	.05

4.) Probability $v > 0$

.15 + .13 + .10 + .10 + .05

= .53

✓Q4

Prob = 53%

5.)

(v)	g(v)	μ	σ^2
0	.47	0	.87
1	.15	.15	.02
2	.13	.26	.05
3	.10	.30	.27
4	.10	.40	.70
5	.05	.25	.66
		1.36	2.57

$\mu = 1.36$ ✓Q5

$\sigma = \sqrt{2.57} = 1.60$

2. Expected Values and Options

Option 25

Price (x)	Probability f(x)	Mean	Variance	Value (v)	g(v)=P(v=v)
20	0.05	1	1.46	0	
21	0.06	1.26	1.17	0	
22	0.07	1.54	0.81	0	
23	0.08	1.84	0.46	0	
24	0.09	2.16	0.18	0	
25	0.12	3	0.02	0	0.47
26	0.15	3.9	0.05	1	0.15
27	0.13	3.51	0.33	2	0.13
28	0.10	2.8	0.67	3	0.10
29	0.10	2.9	1.29	4	0.10
30	0.05	1.5	1.05	5	0.05
	1.00	25.41	7.50		1.00

1)

Mean	**25.41**
Standard Dev	**2.74**

2a)

-1 Standard Dev		+1 Standard Dev
22.67	<= x <=	28.15
Probability =	**0.67**	

2b)

-2 Standard Dev		+2 Standard Dev
19.93	<= x <=	30.89
Probability =	**1.00**	

3)

Value (v)	g(v)=P(v=v)
0	0.47
1	0.15
2	0.13
3	0.10
4	0.10
5	0.05
	1.00

4)

Probability =	0.53

5)

Value (v)	g(v)=P(v=v)	Mean	Variance
0	0.47	0.00	0.87
1	0.15	0.15	0.02
2	0.13	0.26	0.05
3	0.10	0.30	0.27
4	0.10	0.40	0.70
5	0.05	0.25	0.66
	1.00	1.36	2.57

Mean	**1.36**
Standard Dev	**1.60**